# UNIT-10
# Web Security

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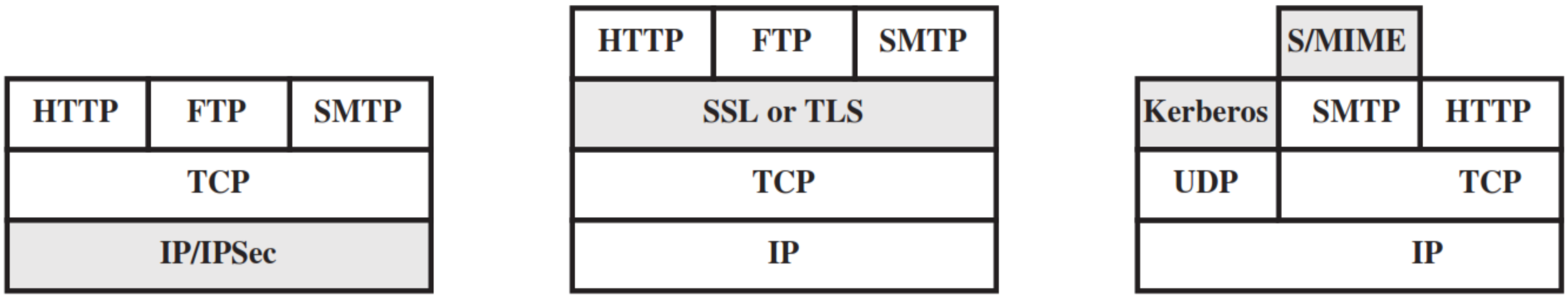
# Outline

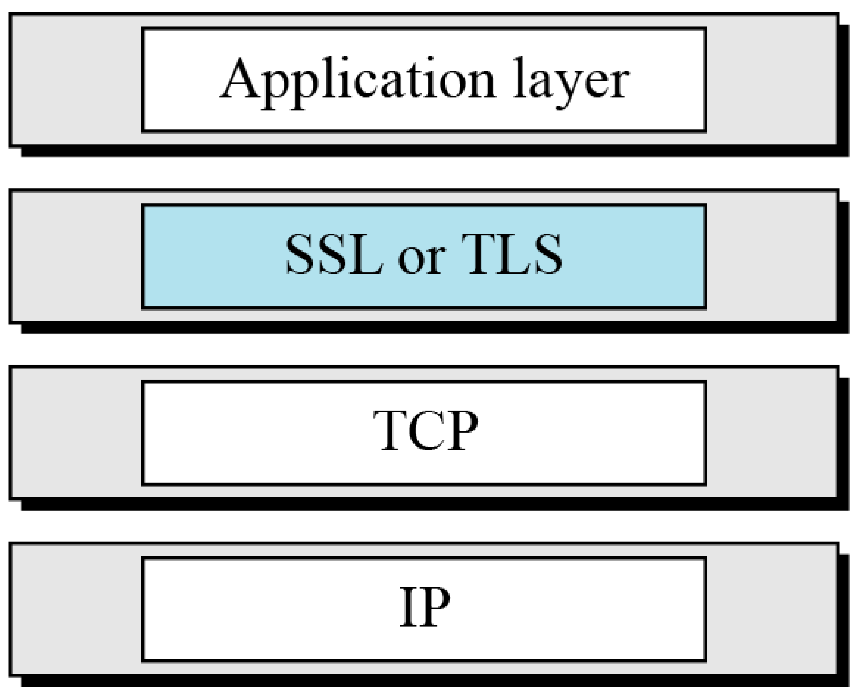
- Web Security threats and approaches
- SSL architecture
- SSL Protocol
- Transport layer security
- HTTPS
- SSH

# Secure Socket Layer (SSL)

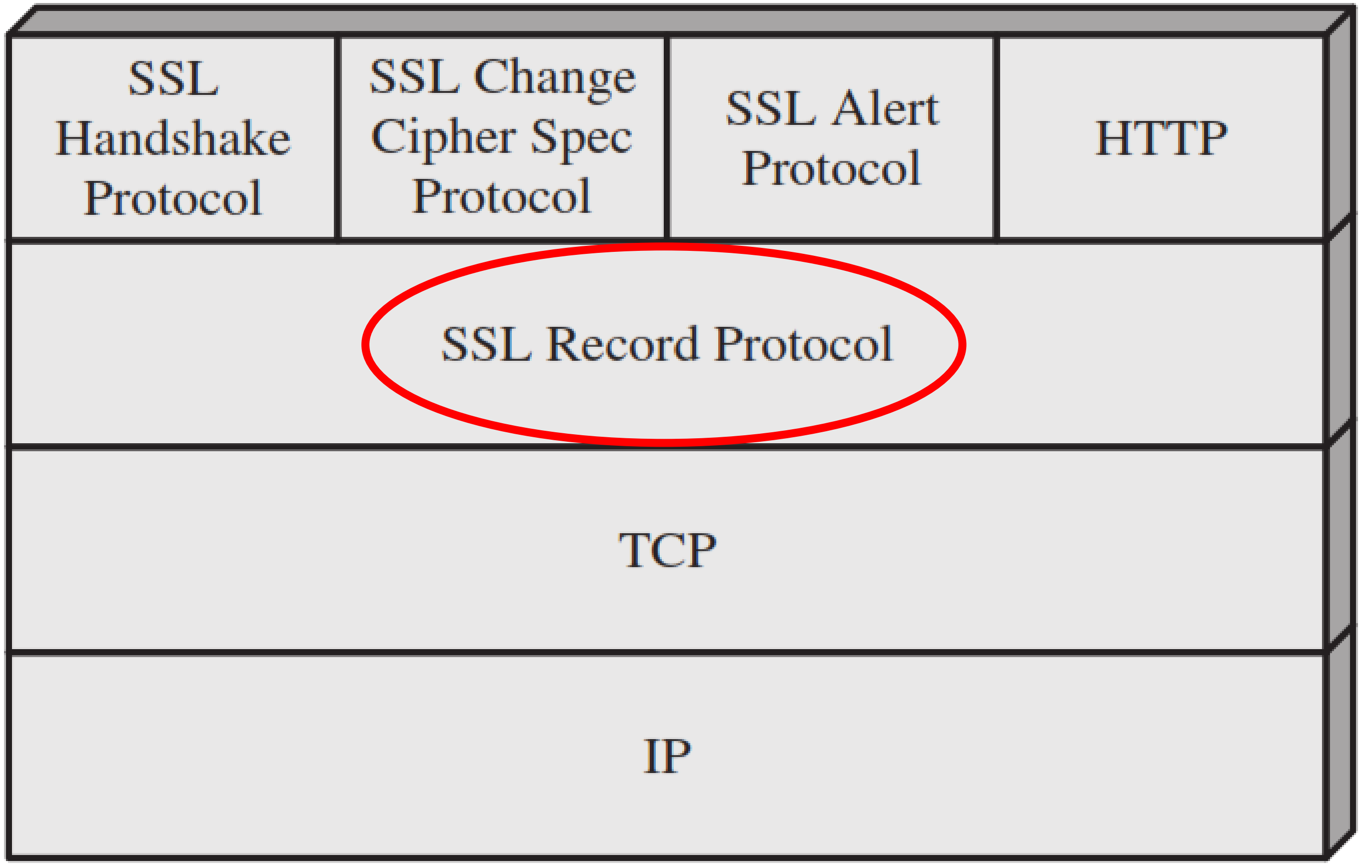
- **Secure Socket Layer (SSL)** provides security services between TCP and applications that use TCP. The Internet standard version is called **Transport Layer Service (TLS)**.
- **SSL/TLS** provides **confidentiality** using symmetric encryption and **message integrity** using a message authentication code.
- **SSL/TLS** includes protocol mechanisms to enable two TCP users to determine the security mechanisms and services they will use.
- **SSL** is designed to make use of TCP to provide a reliable end-to-end secure service.

# Relative Location of Security Facilities in the TCP/IP Protocol Stack

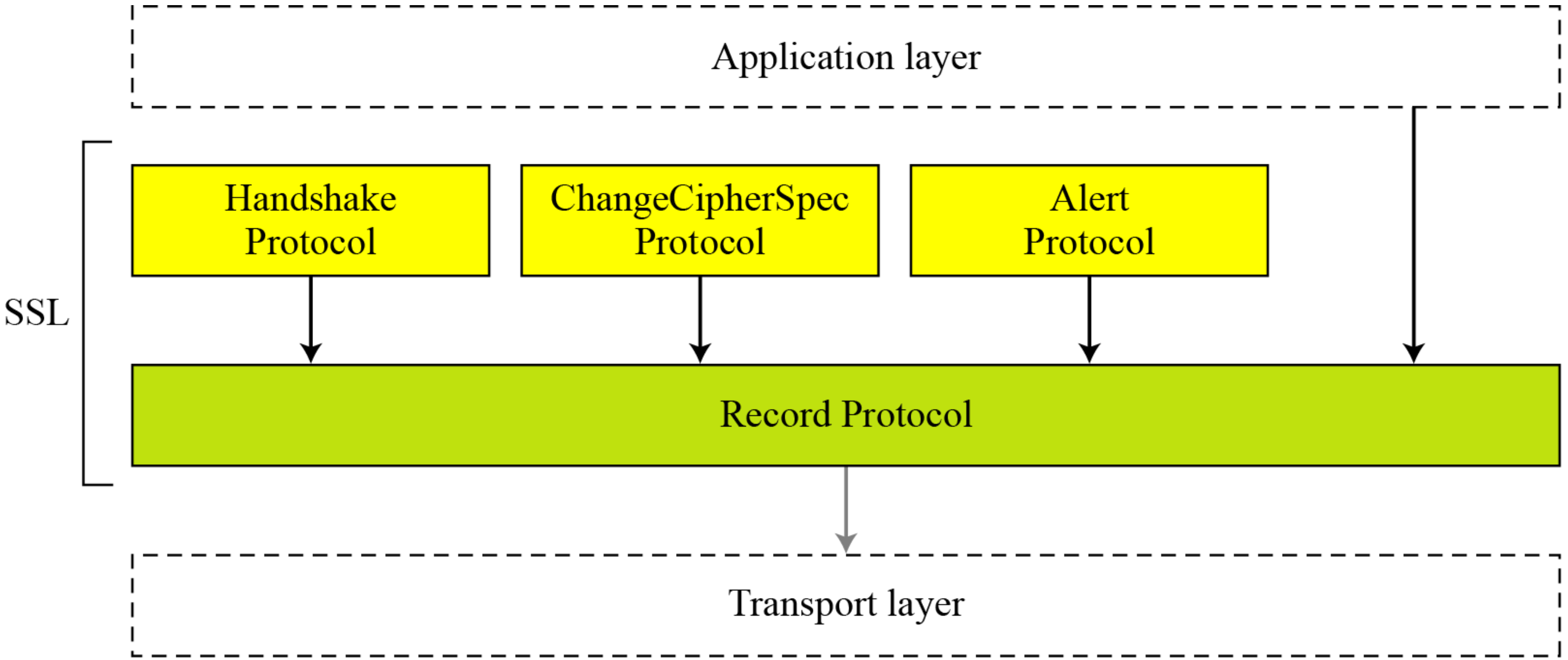


# Secure Socket Layer (SSL) Architecture

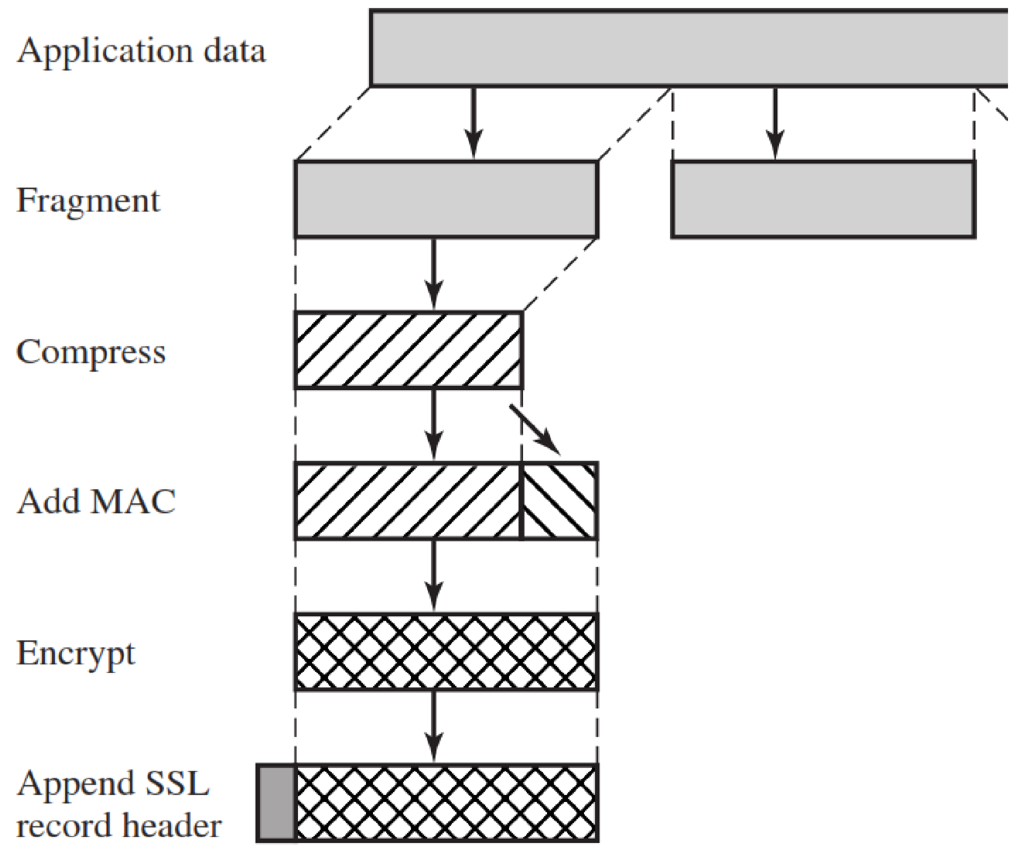


# Four SSL Protocols



# SSL Record Protocol

- It provides two services for SSL connections
- **Confidentiality:** The Handshake Protocol defines a shared secret key that is used for conventional encryption of SSL payloads.
- **Message Integrity:** The Handshake Protocol also defines a shared secret key that is used to form a message authentication code (MAC).



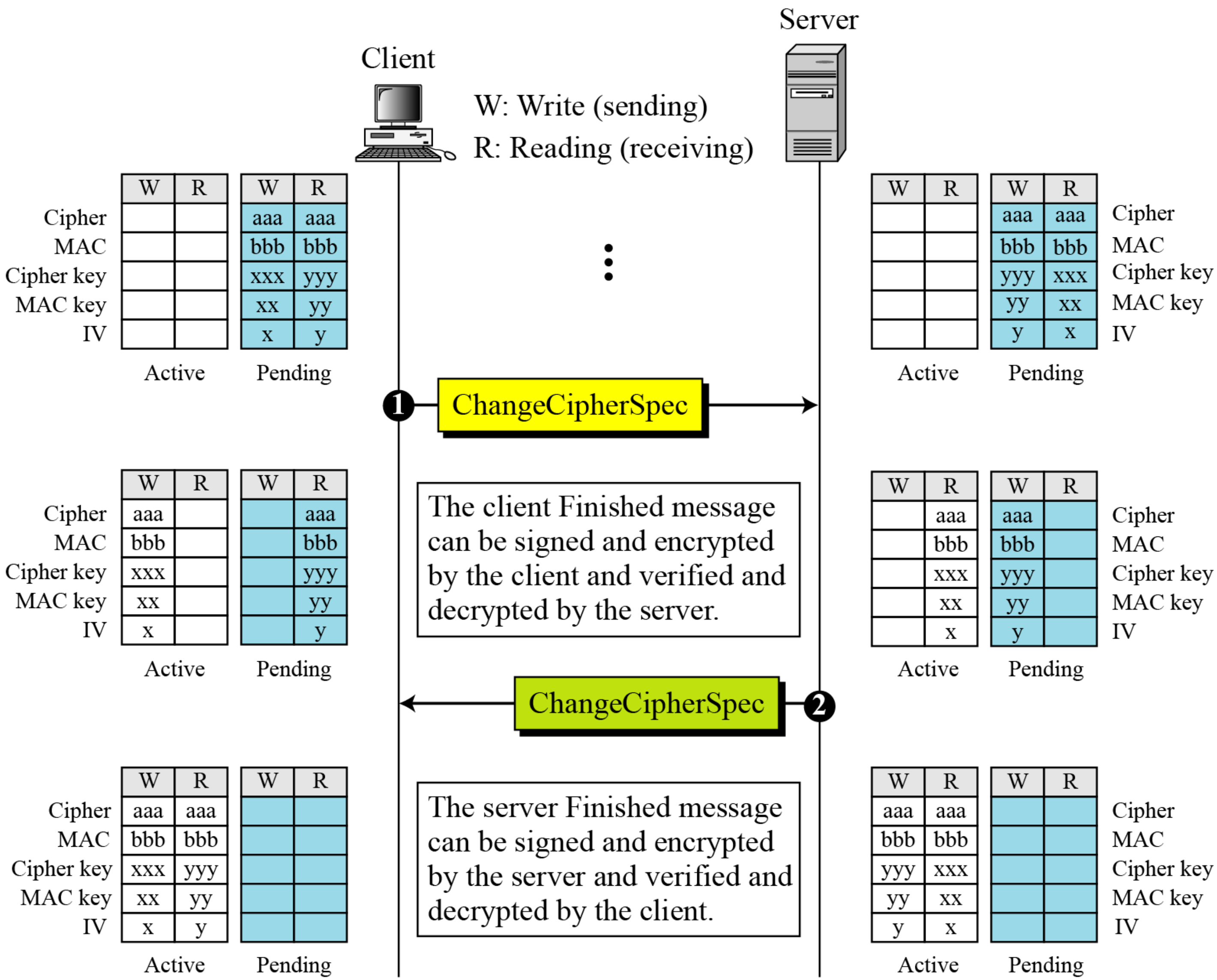
# SSL Record Protocol – Cont…

- The **Record Protocol** takes an application message to be transmitted, fragments the data into manageable blocks, optionally compresses the data, applies a MAC, encrypts, adds a header, and transmits the resulting unit in a TCP segment.
- Received data are decrypted, verified, decompressed, and reassembled before being delivered to higher-level users.

# Change Cipher Spec Protocol

- The **Change Cipher Spec Protocol** is one of the three SSL-specific protocols that use the SSL Record Protocol, and it is the simplest.
- This protocol consists of a **single message** which consists of a single byte with the **value 1**.
- The sole purpose of this message is to cause the pending state to be copied into the current state, which updates the cipher suite to be used on this connection.

# Change Cipher Spec Protocol – Cont…

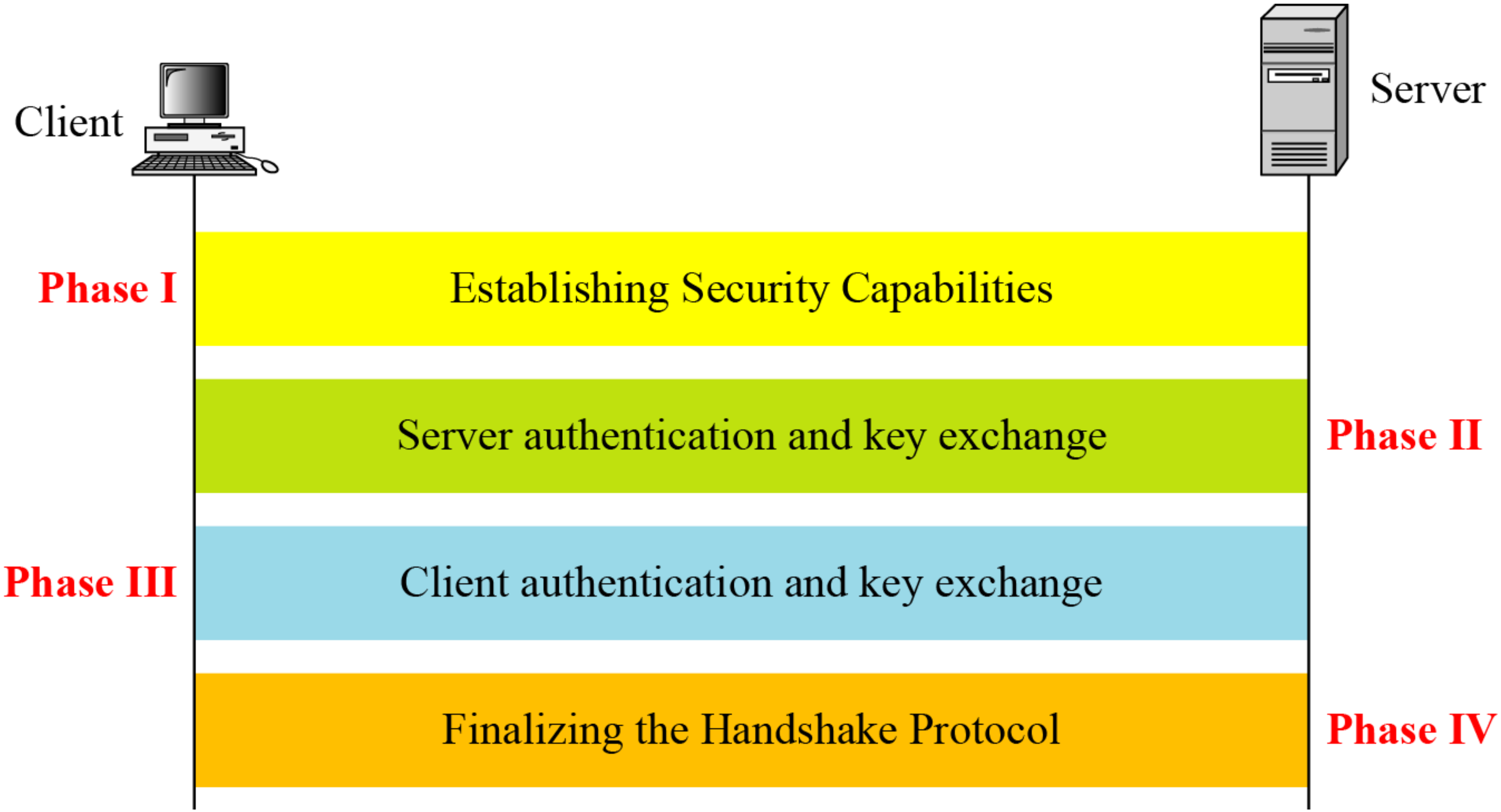


# Alert Protocol

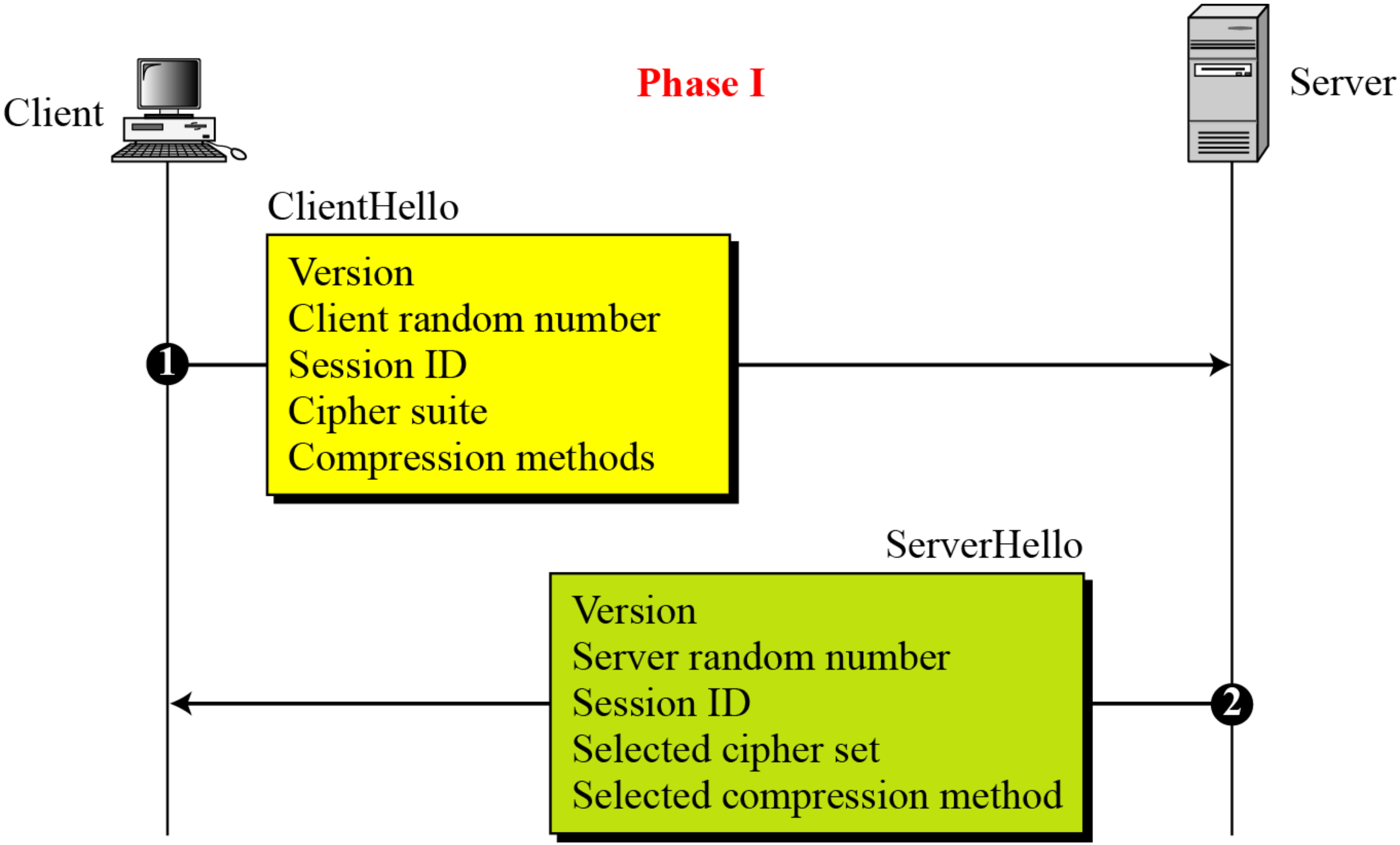
- The **Alert Protocol** is used to convey SSL-related alerts to the peer entity. As with other applications that use SSL, alert messages are compressed and encrypted, as specified by the current state.

| *Value* | *Description* | *Meaning* |
|---|---|---|
| 0 | *CloseNotify* | Sender will not send any more messages. |
| 10 | *UnexpectedMessage* | An inappropriate message received. |
| 20 | *BadRecordMAC* | An incorrect MAC received. |
| 30 | *DecompressionFailure* | Unable to decompress appropriately. |
| 40 | *HandshakeFailure* | Sender unable to finalize the handshake. |
| 41 | *NoCertificate* | Client has no certificate to send. |
| 42 | *BadCertificate* | Received certificate corrupted. |
| 43 | *UnsupportedCertificate* | Type of received certificate is not supported. |
| 44 | *CertificateRevoked* | Signer has revoked the certificate. |
| 45 | *CertificateExpired* | Certificate expired. |
| 46 | *CertificateUnknown* | Certificate unknown. |
| 47 | *IllegalParameter* | An out-of-range or inconsistent field. |

# Handshake Protocol



# Handshake Protocol – Phase I

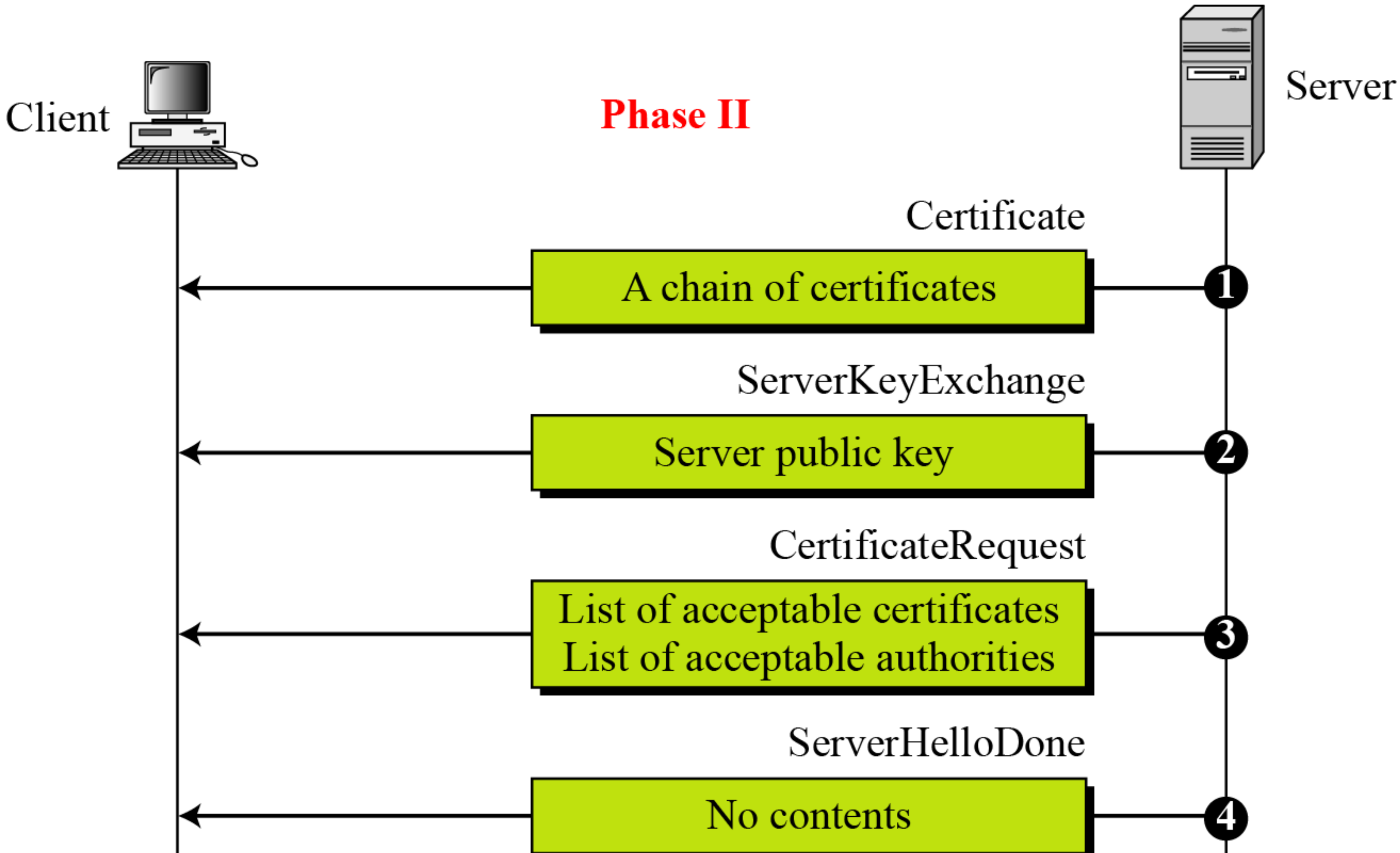


# Handshake Protocol – Phase I

After Phase I, the client and server know the following:

- The **version** of SSL
- The **algorithms** for key exchange, message authentication, and encryption
- The **compression method**
- The two **random numbers** for key generation

# Handshake Protocol – Phase II

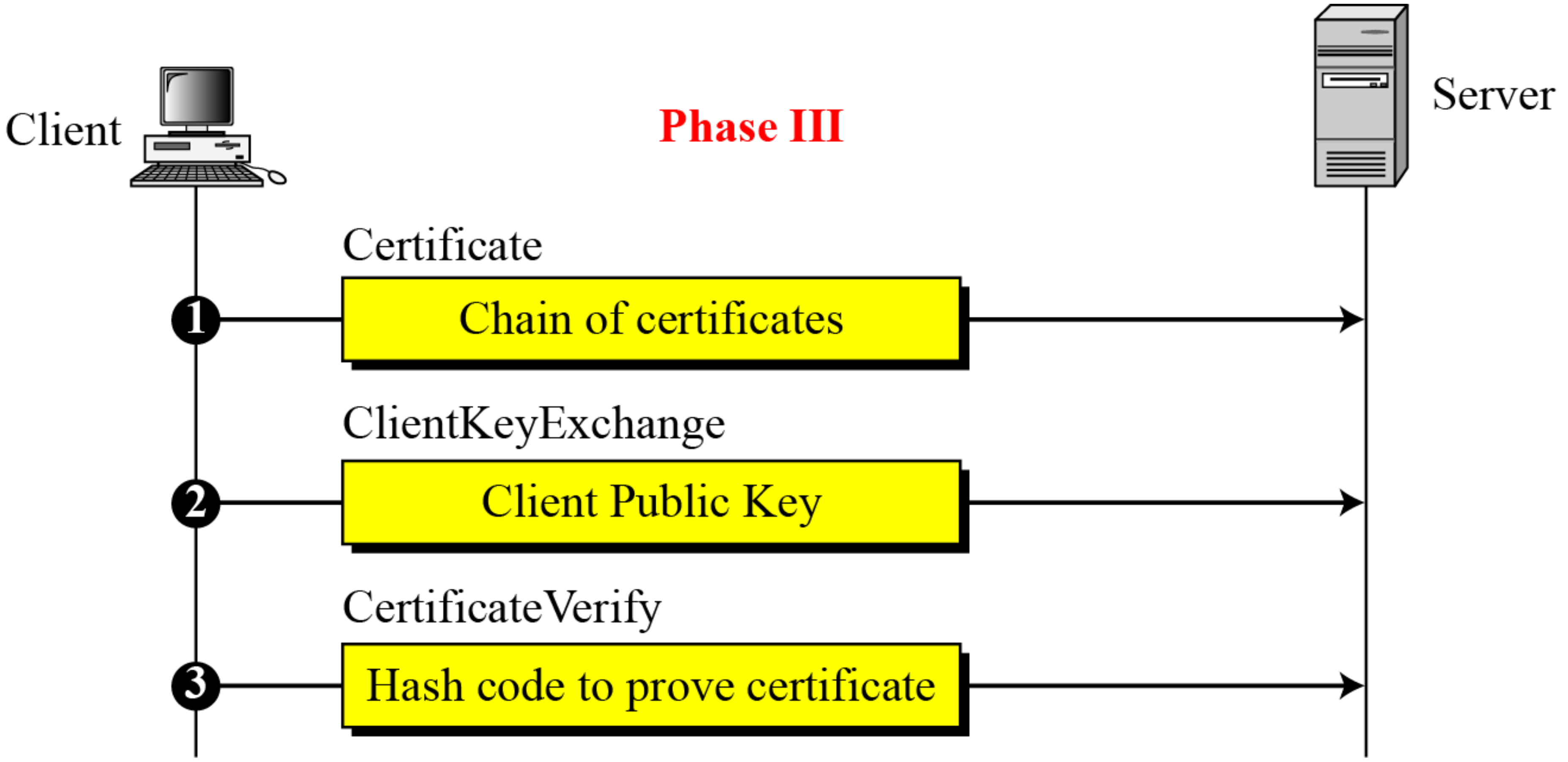


# Handshake Protocol – Phase II

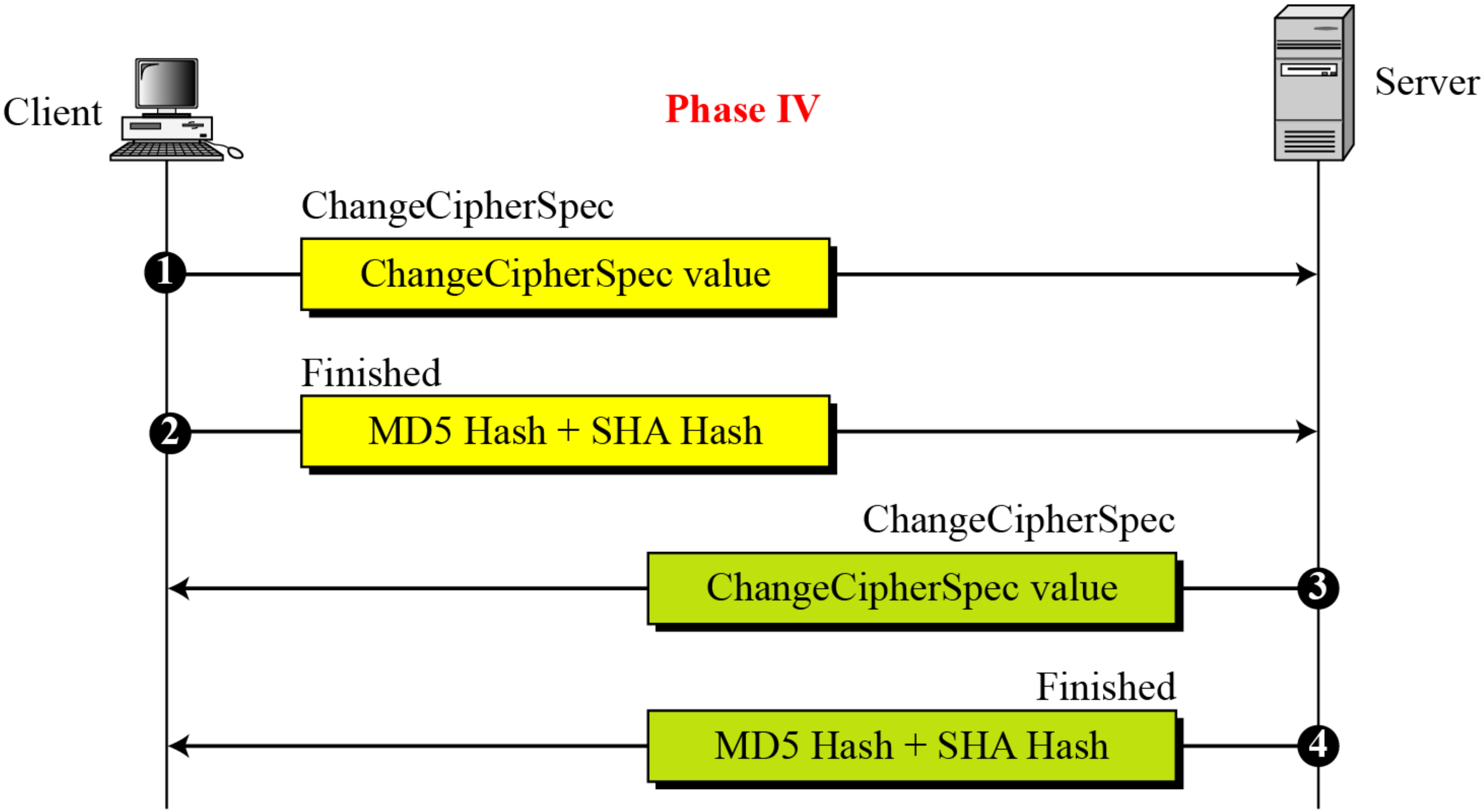
After Phase II

- The server is authenticated to the client.
- The client knows the public key of the server if required.

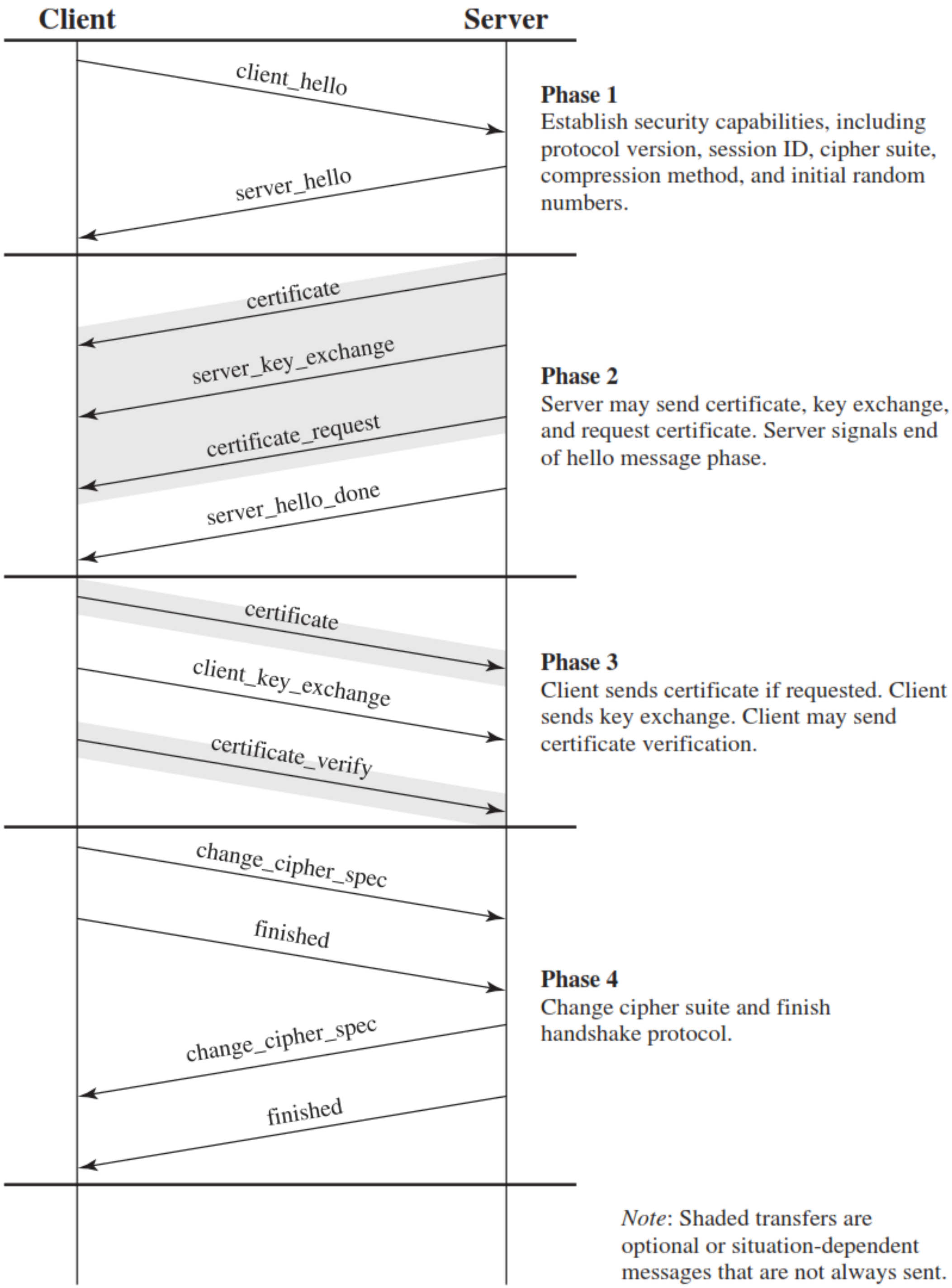
# Handshake Protocol – Phase III



# Handshake Protocol – Phase IV



# SSL Handshake Protocol Phases



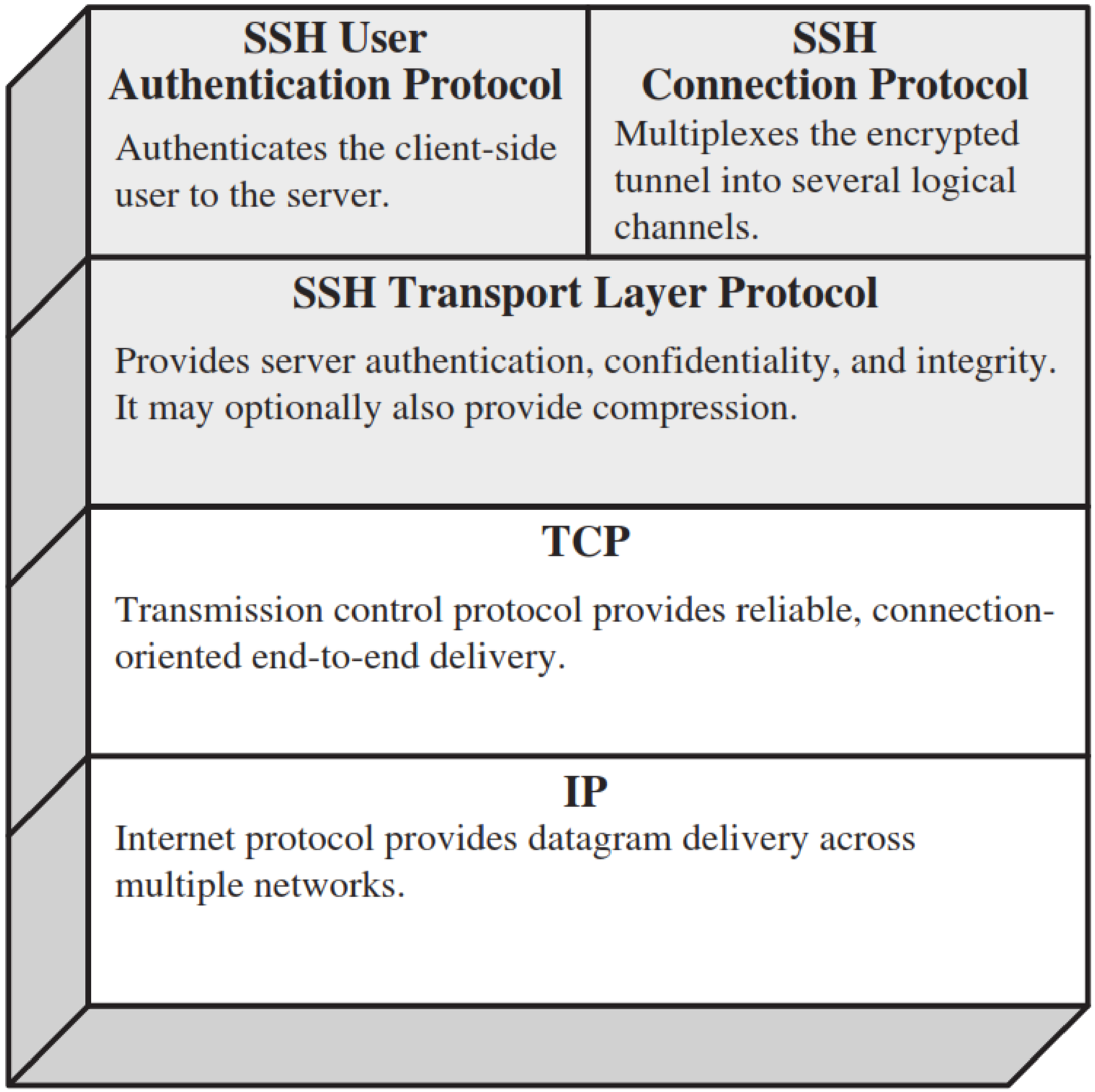
# HTTPS (HTTP over SSL)

- **HTTPS** (HTTP over SSL) refers to the combination of HTTP and SSL to implement secure communication between a Web browser and a Web server.
- When HTTPS is used, the following elements of the communication are encrypted:
  1. **URL** of the requested document
  2. Contents of the **document**
  3. Contents of **browser forms** (filled in by browser user)
  4. **Cookies** sent from browser to server and from server to browser
  5. Contents of **HTTP header**

# SSH (Secure Shell)

- **Secure Shell (SSH)** is a protocol for secure network communications designed to be relatively simple and inexpensive to implement.
- The initial version, SSH1 was focused on providing a **secure remote logon facility** to replace TELNET and other remote logon schemes that provided no security.

# SSH (Secure Shell) – Cont...



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